SPIKE Prime Python Cheat Sheet

How to read this sheet: the `orange line` is the **recipe**. Words in **RED CAPITALS** are blanks you fill in with your own numbers. The blue box under it is a finished example you can copy.

1. Every program looks like this

```
import motor
import motor_pair
import runloop
from hub import port

LEFT = port.A        # left drive motor
RIGHT = port.E       # right drive motor

async def main():
    motor_pair.pair(motor_pair.PAIR_1, LEFT, RIGHT)
    # --- your robot code goes here ---

runloop.run(main())
```

- **Indent with 4 spaces.** Everything inside `main()` is indented.
- Put `await` in front of anything that takes time (moving, waiting, beeping). Forget it and the robot will NOT wait for it to finish.
- `# words after a hashtag` are notes for humans. The robot skips them.
- `print("hi")` shows text on the computer screen. Great for debugging!

2. Our robot's ports

Port	What is plugged in	Write it as
A	Left drive motor (large)	`port.A`
E	Right drive motor (large)	`port.E`
C / D	Attachment motors (medium)	`port.C` `port.D`
B / F	Color sensors	`port.B` `port.F`

3. What the blanks mean

Blank	What it means	Good values
PAIR	The two drive wheels together. Always `motor_pair.PAIR_1`	
PORT	Where the motor or sensor is plugged in	`port.D`
DEGREES	How far the wheel turns. 360 = one full wheel turn = about 17.6 cm. **1 cm ≈ 20 degrees**. Minus = backwards	205 = 10 cm 410 = 20 cm
STEERING	Which way to go. **0 = straight**. 100 = spin right. -100 = spin left. 50 = curve right	0
SPEED	How fast, in degrees per second. Minus = backwards. Max about 1000	200 slow 300 normal 600 fast
MS	Time in milliseconds. **1000 = 1 second**	500 = half sec

4. Driving (both wheels)

First, tell the robot which two motors are the wheels (once, at the top of `main()`):

```
motor_pair.pair(motor_pair.PAIR_1, LEFT, RIGHT)
```

Drive a distance. Straight when STEERING is 0.

```
await motor_pair.move_for_degrees(PAIR, DEGREES, STEERING,
                                  velocity=SPEED)
```

```
# forward 10 cm, straight, normal speed
await motor_pair.move_for_degrees(
    motor_pair.PAIR_1, 205, 0, velocity=300)

# backwards 10 cm  (minus speed = backwards)
await motor_pair.move_for_degrees(
    motor_pair.PAIR_1, 205, 0, velocity=-300)

# spin turn right (100). Use -100 to spin left
await motor_pair.move_for_degrees(
    motor_pair.PAIR_1, 180, 100, velocity=300)
```

Drive for a time instead of a distance.

```
await motor_pair.move_for_time(PAIR, MS, STEERING,
                               velocity=SPEED)
```

```
# drive straight for 2 seconds
await motor_pair.move_for_time(
    motor_pair.PAIR_1, 2000, 0, velocity=300)
```

Start driving and keep going until you say stop. No `await` here!

```
motor_pair.move(PAIR, STEERING, velocity=SPEED)
motor_pair.stop(PAIR)
```

```
motor_pair.move(motor_pair.PAIR_1, 0, velocity=300)
await runloop.sleep_ms(1000)     # keep going 1 second
motor_pair.stop(motor_pair.PAIR_1)
```

5. One motor (arms and attachments)

Careful: for one motor the speed is just the **3rd number**, no `velocity=` word.

```
await motor.run_for_degrees(PORT, DEGREES, SPEED)
```

```
ARM = port.D
# arm up 90 degrees at speed 300
await motor.run_for_degrees(ARM, 90, 300)
# arm down: minus degrees
await motor.run_for_degrees(ARM, -90, 300)
```

```
await motor.run_for_time(PORT, MS, SPEED)
```

```
await motor.run_for_time(ARM, 1000, 300)  # 1 sec
```

```
motor.run(PORT, SPEED)      motor.stop(PORT)
```

```
motor.run(ARM, 300)     # start, keeps going
await runloop.sleep_ms(500)
motor.stop(ARM)
```

6. Waiting and repeating

Wait for some time.

```
await runloop.sleep_ms(MS)
```

```
await runloop.sleep_ms(1000)   # wait 1 second
```

Repeat the indented lines a number of times.

```
for i in range(TIMES):
```

```
for i in range(4):          # do this 4 times
    await motor_pair.move_for_degrees(
        motor_pair.PAIR_1, 360, 0, velocity=300)
    await motor_pair.move_for_degrees(
        motor_pair.PAIR_1, 180, 100, velocity=300)
```

Wait until something is true. The timeout stops it waiting forever.

```
await runloop.until(lambda: SOMETHING IS TRUE, timeout=MS)
```

```
# wait until the hub's left button is pressed
from hub import button
await runloop.until(lambda: button.pressed(button.LEFT))
```

7. Color sensor

```
color_sensor.color(PORT)
   → which color it sees
color_sensor.reflection(PORT)
   → how bright: 0 (black) to 100 (white)
```

```
import color
import color_sensor
EYE = port.B

# decide: is it red?
if color_sensor.color(EYE) == color.RED:
    print("I see red!")
else:
    print("not red")

# drive until the sensor sees black, or 5 seconds pass
motor_pair.move(motor_pair.PAIR_1, 0, velocity=200)
await runloop.until(
    lambda: color_sensor.color(EYE) == color.BLACK,
    timeout=5000)
motor_pair.stop(motor_pair.PAIR_1)
```

Colors you can check for: `color.RED` `color.GREEN` `color.BLUE` `color.YELLOW` `color.BLACK` `color.WHITE`

8. Lights and sounds

```
light_matrix.write("TEXT")
await sound.beep(PITCH, MS)
```

```
from hub import light_matrix, sound

light_matrix.write("Hi!")      # scrolls the words
light_matrix.show_image(light_matrix.IMAGE_HAPPY)
light_matrix.clear()

await sound.beep(440, 300)    # pitch 440, for 300 ms
```

PITCH: bigger number = higher note. 262 = C, 330 = E, 392 = G, 523 = high C.

9. Make your own command

Type the drive command once, then use it with centimeters!

```
async def NAME(BLANKS):
```

```
async def drive_cm(cm, speed=300):
    degrees = int(cm * 20.5)      # cm to degrees
    await motor_pair.move_for_degrees(
        motor_pair.PAIR_1, degrees, 0, velocity=speed)

async def main():
    motor_pair.pair(motor_pair.PAIR_1, LEFT, RIGHT)
    await drive_cm(30)          # forward 30 cm
    await drive_cm(-10, 200)    # back 10 cm, slower
```

10. Uh-oh! Fixing problems

Problem	Try this
Drives backwards	Swap `LEFT` and `RIGHT` ports
Turns the wrong way	Change STEERING 100 to -100 (or back)
Goes too far / not far enough	Change DEGREES. 20 degrees ≈ 1 cm
Does everything at once	You forgot `await`
"no SPIKE hub found"	Hub on? Light blinking? SPIKE app closed?
IndentationError	Fix the spaces at the line start (use 4)
SyntaxError	Missing `:` `)` or `"` on that line or above
NameError	Spelling! Or a missing `import` at the top
Nothing happens, no error	Code inside `main()`? `runloop.run(main())` at bottom?
Robot runs away!	Press the hub's **center button**

💡 **Test small, test often.** Change ONE number, run, watch, write down what happened. Robots are honest: they do exactly what the code says!